NAVAL ARCHITECTURE
INITIAL STABILITY

1 A ship loaded with timber on deck, berths with an angle of loll away from the quay. From which side should the timber on deck be discharged first and why?

2 A vessel or 10 000 tonne displacement has a second moment of area of waterplane about the centerline to be 60 x 10^3 m^4. The centre of buoyancy is 2.75 m above the keel and the following are the disposition of the displacement mass on board.

- 4000 tonne at 6.30 m above the keel
- 2000 tonne at 7.50 m above the keel
- 4000 tonne at 9.15 m above the keel

Calculate the metacentric height.

[2.22 m]

3 A vessel of constant rectangular cross-section is 7.2 m wide.

a. Draw the metacentric diagram for the vessel using 0.5 m intervals of draught up to the 4.0 m waterline.
b. If the centre of gravity is 3.0 m above the keel, determine from the diagram the limits of draught between which the vessel will be unstable.

[2.25 m to 3.64 m]

4 A ship of 12 000 tonne displacement and 15 m beam has a metacentric height of 1.1 m. A mass 80 tonne is lifted from its position on the centreline at the bottom of the lower hold by one of the ship's derricks, and placed on the quay 2.0 m from the ship's side. The ship heels to a maximum of 3.5º when the mass is being moved.

a. Calculate the height of the derrick head above the centre of gravity of the mass.
b. By how much does the GM_T alter during the operation?

[9.853 m; 0.066 m reduction]

5 A ship of 4000 tonne displacement has the centre of gravity 7.3 m above the keel. Loads 800, 600 and 400 tonnes are added at 3.7 m, 4.3 m and 6.7 m respectively above the keel, while a load of 300 tonne was discharged from a position 0.6 m above the keel.

Determine the metacentric height in the final condition and the moment of statical stability at a heel of 8º, using data extracted from the attached curves, given that the mean draught is 7.1 m.

[2.93 m; 558.78 tonne.m]

6 A crane in a cargo ship of 5100 tonne displacement and GM_T 1 m, lifts a weight of 22 tonne from a cargo hold through a vertical height of 10 m. If the height of the crane's jib is 20 m above the hold and the horizontal radius of the jib is 10 m, calculate the reduction in metacentric height caused by lifting of the weight.

What is the approximate angle of heel caused by the turning the crane through 30 degrees assuming it is initially aligned fore and aft?